Statistics for International Commerce

Week 14: Multiple regression analysis

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Agenda

- multiple predictors
- interpretation of coefficients
- prediction
- model fit
- business applications

Learning Objectives

After this lecture, you should be able to:

- explain multiple regression
- interpret regression coefficients
- understand ceteris paribus interpretation
- evaluate model fit
- identify limitations of regression models

Motivation

A company wants to predict:

customer spending

using:

- income
- advertising exposure
- delivery speed

> One variable alone may not be enough.

From Simple to Multiple Regression

Simple Regression

$$Y = a + bX$$

uses:

- one independent variable

Multiple Regression

$$Y = a + b_1X_1 + b_2X_2 + b_3X_3$$

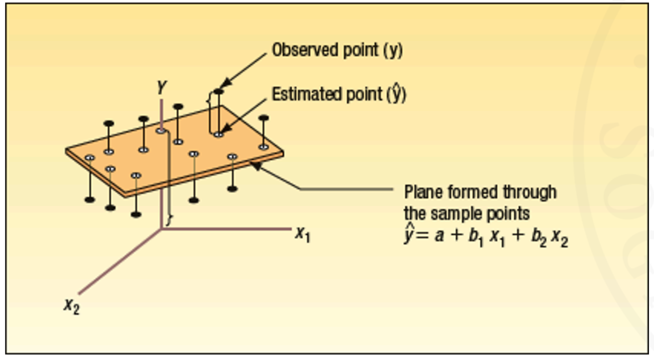
uses:

- multiple independent variables

Multiple Regression Analysis

The general multiple regression equation with k independent variables is given by:

GENERAL MULTIPLE REGRESSION EQUATION $\hat{y} = a + b_1x_1 + b_2x_2 + b_3x_3 + \cdots + b_kx_k$ **[14–1]**



In the model:
$Y = a + b_1X_1 + b_2X_2 + \cdots + b_kX_k$:

- $X_1, \ldots, X_k$ are independent variables
- a is the intercept (predicted Y when all X values are 0)
- b_i is the expected change in Y for a one-unit increase in X_i, holding all other variables constant

Each b_i is called a partial regression coefficient.

The coefficients are estimated using the least squares method (typically with software such as Excel, MINITAB, or Python).

Why Use Multiple Regression?

Real business outcomes usually depend on:

- many factors simultaneously

Examples:

- sales
- customer satisfaction
- stock prices
- shipping costs

Example Business Question

What affects online sales?

Possible factors:

- advertising
- price
- delivery time
- customer ratings

Example Dataset

```
##         sales  advertising  delivery_days     income
## 0  187.202652    35.599575       2.855113  58.594221
## 1  259.456701    42.368062       6.577754  78.588341
## 2  236.979998    29.000020       3.309519  78.679752
## 3  243.847912    57.192601       4.927591  55.863373
## 4  176.618908    44.672124       3.453454  38.390964
```

Variables

Variable	Meaning	units
sales	monthly sales	dollars
advertising	advertising spending	dollars
delivery_days	average delivery time	days
income	customer income	dollars

Correlation Matrix

```
##                sales  advertising  delivery_days  income
## sales           1.00         0.57           0.08    0.85
## advertising     0.57         1.00           0.19    0.22
## delivery_days   0.08         0.19           1.00    0.12
## income          0.85         0.22           0.12    1.00
```

Multiple Regression Equation

$$Sales = a + b_1 Advertising + b_2 Delivery + b_3 Income$$

Important Idea

Each coefficient measures:

> the effect of one variable while holding other variables constant

This is called:

ceteris paribus

Estimating the Model

```
##          Variable Coefficient Robust SE  t-stat p-value
## 0     Advertising     1.789***   (0.175)  10.242   0.000
## 1   Delivery days     -3.581*    (1.980)  -1.809   0.071
## 2          Income     1.994***   (0.111)  18.009   0.000
## 3        Constant    39.686***  (11.556)   3.434   0.001
```

```
##          Statistic  Value
## 0     Observations     60
## 1        R-squared  0.881
## 2   Adj. R-squared  0.875
```

```
## Notes: Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.10
```

Important Output

Focus on:

- coefficients
- robust standard errors
- t-statistics and p-values
- significance stars
- R-squared

Do not panic about all other statistics.

Interpreting Coefficients

Suppose:

$$b_1 = 2.5$$

Interpretation:

Holding other variables constant, one additional unit of advertising increases expected sales by 2.5 units.

Negative Coefficients

Suppose:

$$b_2 = -4.9$$

Interpretation:

Holding other variables constant, one additional delivery day reduces expected sales by 4.9 units.

Why "Holding Other Variables Constant" Matters

Without controlling for other variables:

- relationships may become misleading

Example:

- wealthier customers may both:
    - spend more
    - receive faster delivery

Evaluating the Model: Coefficient of Determination

Suppose:

$$R^2 = 0.804$$

Interpretation:

- 80.4% of the variation in sales is explained by the independent variables in the model
- the remaining 19.6% is due to random variation and other factors not included in the model

Higher R^2 usually indicates a better fit, but it does not guarantee that the model is correct.

COEFFICIENT OF MULTIPLE DETERMINATION

$$R^2 = \frac{\text{SSR}}{\text{SS total}} \qquad [14\text{–}3]$$

Source	df	SS	MS	F
Regression	k	SSR	MSR = SSR/k	MSR/MSE
Residual or error	$n-(k+1)$	SSE	MSE = SSE/$[n-(k+1)]$	
Total	$n-1$	SS total		

ANOVA

	df	SS	MS	F	Significance F
Regression	3	171220.473	57073.491	21.901	0.000
Residual	16	41695.277	2605.955		
Total	19	212915.750			

$$R^2 = \frac{\text{SSR}}{\text{SS total}} = \frac{171{,}220.473}{212{,}915.750} = .804$$

Adjusted Coefficient of Determination

Adding more independent variables almost always increases R^2.

That creates a problem:

- a larger R^2 does not always mean a better model
- some added variables may contribute very little useful information

Adjusted R^2 corrects for this by penalizing unnecessary predictors.

Use adjusted R^2 when comparing models with different numbers of independent variables.

Its formula accounts for both sample size and degrees of freedom.

ADJUSTED COEFFICIENT OF DETERMINATION

$$R^2_{\text{adj}} = 1 - \frac{\dfrac{\text{SSE}}{n-(k+1)}}{\dfrac{\text{SS total}}{n-1}} \qquad [14\text{–}4]$$

Prediction Example

Predict sales for:

Variable	Value
advertising	60
delivery_days	3
income	80

```
## Predicted monthly sales: 295.80

## 95% prediction interval: [267.14, 324.46]

## Interpretation: For a customer profile with advertising=60, delivery_days=3, and income=80,

## the model predicts expected sales near the point estimate above.
```

Residuals

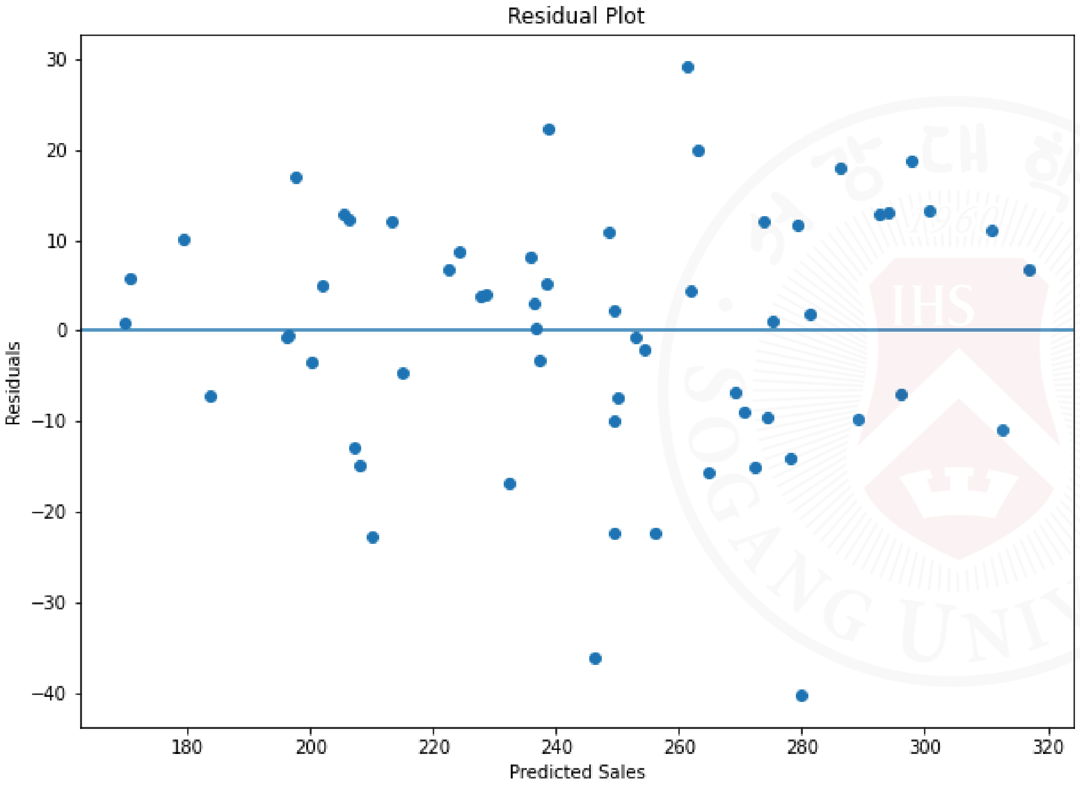
Residual:

$$e = y - \hat{y}$$

Meaning:

- prediction error

Residual Plot



What We Want

Residuals should look:

- random
- patternless

Patterns may indicate:

- missing variables
- nonlinear relationship
- poor model fit

Multicollinearity

Problem:

Independent variables may be strongly correlated with each other.

Example:

- income
- education
- wealth

Example

If:

- advertising spending
- influencer spending

move together strongly, it becomes difficult to separate their effects.

Why Multicollinearity Is a Problem

It may:

- distort coefficient estimates
- increase uncertainty
- make interpretation difficult

Correlation Among Predictors

```
##                advertising  delivery_days  income
## advertising           1.00           0.19    0.22
## delivery_days         0.19           1.00    0.12
## income                0.22           0.12    1.00
```

The Global Test of the Model

The global F-test checks whether the model, as a whole, has explanatory power.

Step 1: State the hypotheses.

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_k = 0$$

$$H_1 : \text{At least one } \beta_i \neq 0$$

Step 2: Choose a significance level.

$$\alpha = 0.05$$

Step 3: Use the F-statistic with k and $n - (k + 1)$ degrees of freedom.

Step 4: Decision rule: reject H_0 if the test statistic is larger than the critical F value.

$$H_0 : \beta_1 = \beta_2 = \ldots = \beta_k = 0$$

$$H_1 : \text{ Not all } \beta \text{ s equal } 0$$

Decision rule:

Reject H_0 if $F > F_{\alpha,k,n-(k+1)}$

The Global Test of the Model: Interpretation

Step 5: Estimate the model and compute the F-statistic.

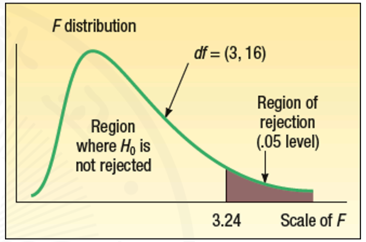
SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.897
R Square	0.804
Adjusted R Square	0.767
Standard Error	51.049
Observations	20

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	3	171220.473	57073.491	21.901	0.000
Residual	16	41695.277	2605.955		
Total	19	212915.750			

	Coefficients	*Standard Error*	*t Stat*	*P-value*
Intercept	427.194	59.601	7.168	0.000
Temp	-4.583	0.772	-5.934	0.000
Insul	-14.831	4.754	-3.119	0.007
Age	6.101	4.012	1.521	0.148



GLOBAL TEST $$F = \frac{SSR/k}{SSE/[n-(k+1)]} \quad [14\text{–}5]$$

$$F = \frac{SSR/k}{SSE/[n-(k+1)]} = \frac{171{,}220.473/3}{41{,}695.277/[20-(3+1)]} = 21.90$$

If the p-value is less than α, reject H_0.

Interpretation:

- the regression model is statistically significant overall
- at least one independent variable helps explain variation in the dependent variable
- this does not mean that every coefficient is individually significant

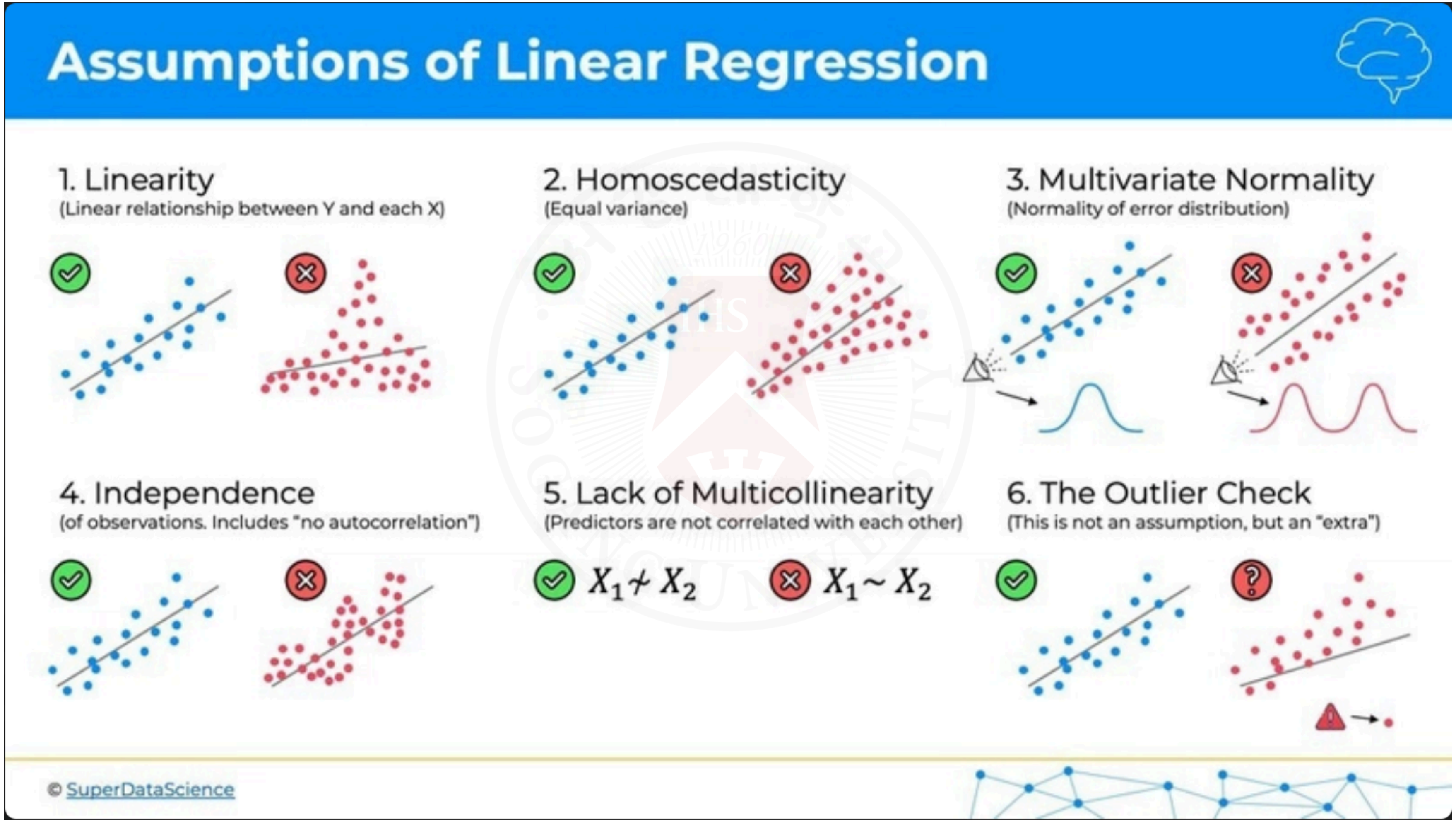
Evaluating the Assumptions

Key assumptions of multiple regression:

- linearity: the relationship between Y and the predictors is linear in parameters
- constant variance: the error variance is stable across observations
- normality: the error terms are approximately normally distributed
- no perfect multicollinearity: predictors are not exact linear combinations of one another
- independence: the error terms are independent

If these assumptions are badly violated, interpretation and prediction become less reliable.

Evaluating the Assumptions (cont.)



Evaluating the Assumptions: Multicollinearity

Multicollinearity occurs when independent variables are strongly correlated with each other.

Possible warning signs:

- an important predictor appears statistically insignificant
- coefficient estimates change sharply when a variable is added or removed
- standard errors become large, making interpretation unstable

Checking Multicollinearity

Two common checks are:

- pairwise correlations among predictors
- the variance inflation factor (VIF)

Rules of thumb:

- correlations between about -0.70 and 0.70 are often not a major concern
- a VIF greater than 10 suggests serious multicollinearity

To compute the VIF for one predictor, regress that predictor on the remaining predictors and use the resulting R^2.

Multicollinearity Example: Correlation Check

In the home-heating example, the predictors are:

- outside temperature
- insulation thickness
- furnace age

First, examine the correlation matrix for the independent variables.

	Cost	*Temp*	*Insul*	*Age*
Cost	1.000			
Temp	-0.812	1.000		
Insul	-0.257	-0.103	1.000	
Age	0.537	-0.486	0.064	1.000

Because none of the pairwise correlations are close to 0.70 or -0.70, multicollinearity does not appear to be severe.

Multicollinearity Example: VIF Check

To examine multicollinearity more carefully, compute the VIF for temperature.

Regress temperature on the other predictors:

$$\text{Temperature} = b_0 + b_1(\text{Insulation}) + b_2(\text{Furnace Age})$$

SUMMARY OUTPUT					
Regression Statistics					
Multiple R	0.491				
R Square	0.241				
Adjusted R Square	0.152				
Standard Error	16.031				
Observations	20				
ANOVA					
	df	*SS*	*MS*	*F*	*Significance F*
Regression	2	1390.291	695.145	2.705	0.096
Residual	17	4368.909	256.995		
Total	19	5759.200			

$$VIF = \frac{1}{1-R_j^2} = \frac{1}{1-0.241} = \frac{1}{.759} = 1.32$$

Because the VIF is 1.32, which is well below 10, temperature is not highly collinear with the other predictors.

Conclusion: multicollinearity is not a serious problem in this example.

Assessing Studies Based on Multiple Regression

What makes a multiple regression study reliable?

Internal validity:

- Are the estimated causal effects valid for the population studied?

External validity:

- Can the findings be generalized to other populations and settings?

Threats to External Validity

Common threats include:

- population differences: results from one group may not apply to another
- setting differences: institutions, laws, and environments may differ

Example:

- lab evidence from mice may not directly generalize to humans

Consistent findings across multiple high-quality studies strengthen external validity.

Threats to Internal Validity

Key threats in multiple regression:

- omitted variable bias
- simultaneity (reverse causality)
- functional form misspecification

Internal Validity Threat 1: Omitted Variable Bias

Omitted variable bias occurs when an omitted factor:

- affects the dependent variable Y
- is correlated with an included regressor

Then the estimated coefficient mixes the true effect with the omitted factor's effect.

Omitted Variable Bias Example

Claim: "Golfers are more prone to heart disease, cancer, and arthritis."

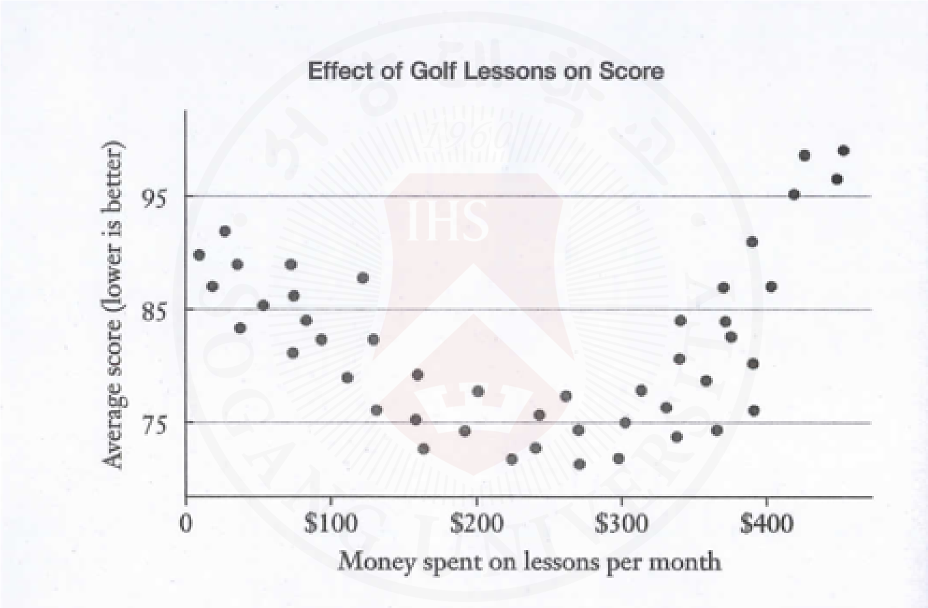
Possible omitted variable: age.

- older people are more likely to play golf
- older people also have higher health risks

Without controlling for age, golf may be incorrectly blamed.

Internal Validity Threat 2: Functional Form Misspecification

If the true relationship is nonlinear but we estimate a linear model, OLS estimates can be biased.



Common fixes:

- polynomial terms (e.g., X^2, X^3)
- logarithmic transformations of Y and/or X

Internal Validity Threat 3: Simultaneity

Simultaneity (reverse causality) occurs when causality runs both ways.

Examples:

- police presence and crime
- poor golf performance and number of lessons

This makes it difficult to isolate one-direction causal effects.

Common fixes:

- instrumental variables
- natural experiments

Overcoming Internal Validity Threats

Key challenge:

- causal inference needs a counterfactual (what would have happened without treatment)

But we never observe both states for the same unit at the same time.

Researchers use study designs that approximate the missing counterfactual.

Example: Police and Crime

Naive regression can be misleading:

- more police may reduce crime
- higher crime may also lead to hiring more police

A useful strategy is to find exogenous variation.

Example: high terrorism alert days in Washington, D.C., which increased police presence for reasons unrelated to ordinary street crime.

Design Strategies

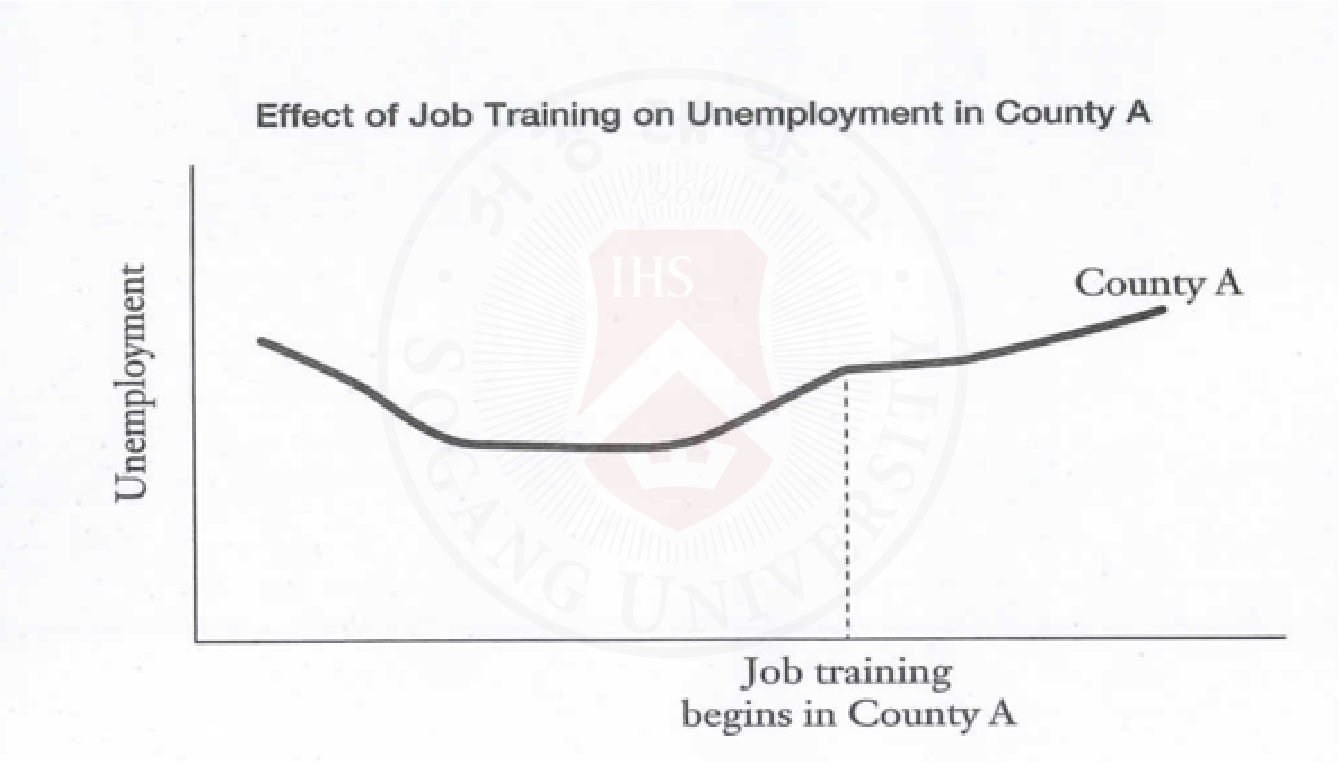
1. Randomized controlled trials (RCTs)
2. Natural experiments
3. Difference-in-differences (DiD)

Natural experiment example:

- changes in compulsory schooling laws across states can identify education effects on health outcomes

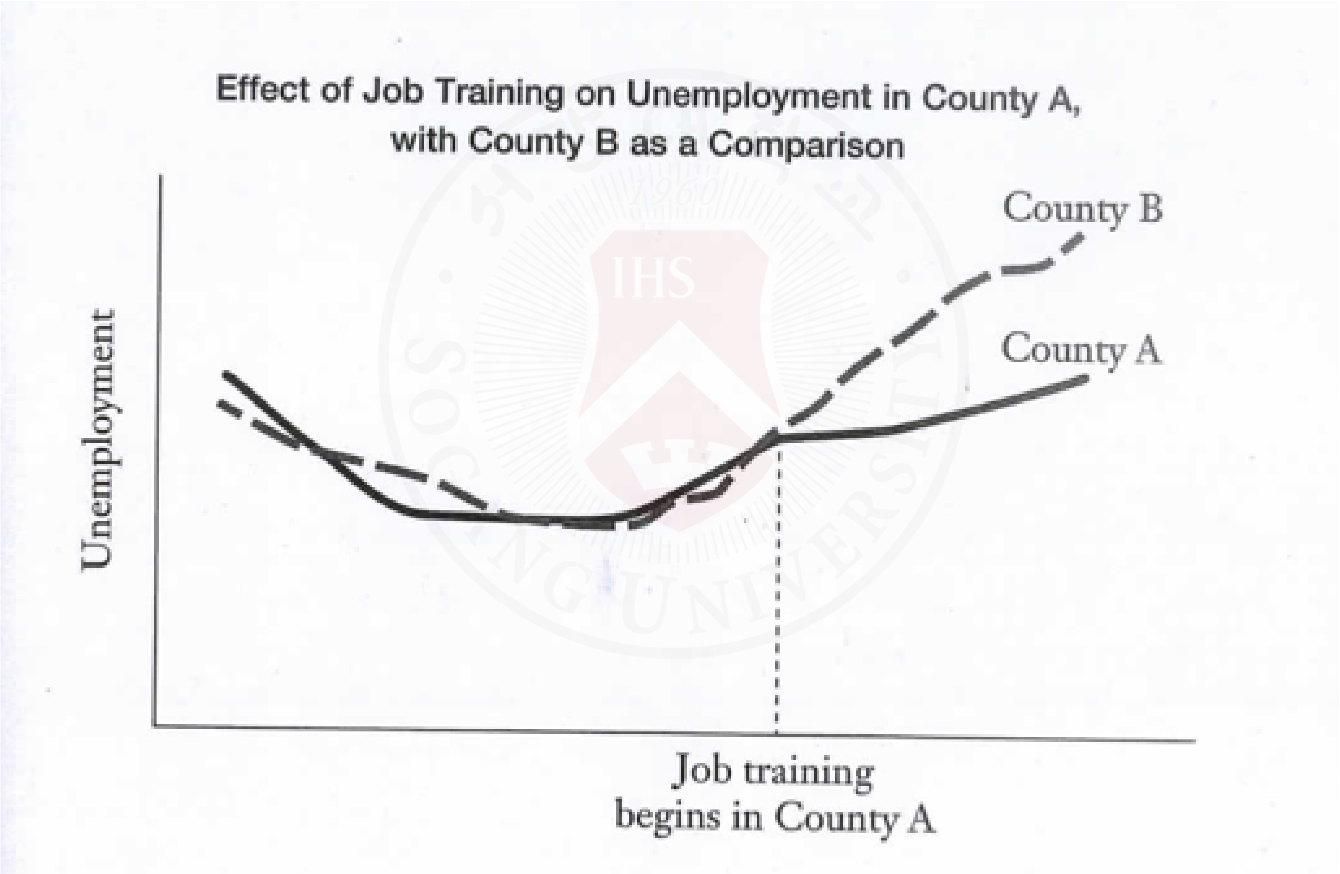
Difference-in-Differences: Step 1

Compare the treated group before and after the policy.



Difference-in-Differences: Step 2

Compare that change with a similar untreated group over the same period.



Business Applications

Multiple regression is widely used for:

- sales forecasting
- pricing analysis
- customer analytics
- demand estimation
- credit scoring
- logistics optimization

Example Questions

- What affects customer spending?
- What determines delivery satisfaction?
- Which variables affect exports?
- Which factors predict firm profitability?

Important Warning

Regression does NOT automatically prove causation.

Even with multiple variables:

- omitted variables may still exist
- reverse causality may exist

Common Mistakes

- interpreting correlation as causation
- ignoring multicollinearity
- overfitting
- using too many irrelevant variables

Simple vs Multiple Regression

Simple	Multiple
one predictor	several predictors
easier interpretation	more realistic
limited control	controls for more factors

In-Class Practice

See Week 14 workbook in Cyber Campus folder for hands-on practice with multiple regression analysis using Python.

Final Idea

Multiple regression helps businesses understand:

which factors matter most

while controlling for other variables.

One Sentence Summary

Multiple regression models the relationship between one outcome and several explanatory variables simultaneously.

Remaining Schedule

(June 2/4) Multiple regression analysis (LMW Chapter 14)

- Lab: Multiple regression in Python

(June 9) Multiple Time Series and Forecasting (LMW Chapter 16)

- Lab: Time series forecasting in Python
- Final review and Q&A session

(June 11) Project presentations

(June 16) No class (final exam week)

(June 18) Final Exam

Any questions?

Thank you for your attention and active participation!